MAT2002 HW4 文杰 123090612

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1. Verification of Solutions

- **For $y_1 = 1$:**
  - $y_1' = 0, y_1'' = 0, y_1''' = 0$.
  - Substitution: $x(0) - 0 = 0$. **(Verified)**
- **For $y_2 = x$:**
  - $y_2' = 1, y_2'' = 0, y_2''' = 0$.
  - Substitution: $x(0) - 0 = 0$. **(Verified)**
- **For $y_3 = x^3$:**
  - $y_3' = 3x^2, y_3'' = 6x, y_3''' = 6$.
  - Substitution: $x(6) - (6x) = 6x - 6x = 0$. **(Verified)**

2. Determination of the Wronskian

According to **Definition 8.5**:

$$W(1, x, x^3) = \det \begin{pmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{pmatrix} = \det \begin{pmatrix} 1 & x & x^3 \\ 0 & 1 & 3x^2 \\ 0 & 0 & 6x \end{pmatrix}$$

Since this is an upper triangular matrix, the determinant is the product of the diagonal elements:

$$W(x) = 1 \cdot 1 \cdot 6x = 6x$$

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(i) $y''' - 5y'' + 3y' + y = 0$

Characteristic equation:

$$r^3 - 5r^2 + 3r + 1 = 0.$$

Factor:

$$r^3 - 5r^2 + 3r + 1 = (r - 1)(r^2 - 4r - 1).$$

So the roots are

$$r = 1, \qquad r = 2 + \sqrt{5}, \qquad r = 2 - \sqrt{5}.$$

These are three distinct real roots, so the general solution is

$$\boxed{y(t) = C_1 e^t + C_2 e^{(2+\sqrt{5})t} + C_3 e^{(2-\sqrt{5})t}}$$

(ii) $y''' + 5y'' + 6y' + 2y = 0$

Characteristic equation:

$$r^3 + 5r^2 + 6r + 2 = 0.$$

Factor:

$$r^3 + 5r^2 + 6r + 2 = (r + 1)(r^2 + 4r + 2).$$

So the roots are

$$r = -1, \qquad r = -2 + \sqrt{2}, \qquad r = -2 - \sqrt{2}.$$

Again these are three distinct real roots, hence the general solution is

$$\boxed{y(t) = C_1 e^{-t} + C_2 e^{(-2+\sqrt{2})t} + C_3 e^{(-2-\sqrt{2})t}}$$

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(i) $y^{(4)} - 4y'' = t^2 + e^t$

Consider first the homogeneous equation

$$y^{(4)} - 4y'' = 0.$$

Its characteristic equation is

$$r^4 - 4r^2 = 0 \quad \Longrightarrow \quad r^2(r^2 - 4) = 0 \quad \Longrightarrow \quad r = 0, 0, 2, -2.$$

Since $r = 0$ is a repeated root of multiplicity 2, the corresponding homogeneous solution is

$$y_c(t) = c_1 + c_2 t + c_3 e^{2t} + c_4 e^{-2t}.$$

For

$$y^{(4)} - 4y'' = t^2 + e^t,$$

Split it as

$$Y = Y_1 + Y_2,$$

where

$$y^{(4)} - 4y'' = t^2 \quad \text{and} \quad y^{(4)} - 4y'' = e^t.$$

For the polynomial part, since $r = 0$ is already a double root of the characteristic equation, multiply the usual quadratic trial polynomial by t^2. So try

$$Y_1 = t^2(At^2 + Bt + C) = At^4 + Bt^3 + Ct^2.$$

Then

$$Y_1'' = 12At^2 + 6Bt + 2C, \qquad Y_1^{(4)} = 24A.$$

Hence

$$Y_1^{(4)} - 4Y_1'' = 24A - 4(12At^2 + 6Bt + 2C) = -48At^2 - 24Bt + (24A - 8C).$$

Match coefficients with t^2:

$$\begin{cases} -48A = 1 \\ -24B = 0 \\ 24A - 8C = 0 \end{cases} \Longrightarrow \begin{cases} A = -\frac{1}{48} \\ B = 0 \\ C = -\frac{1}{16} \end{cases} \Longrightarrow Y_1 = -\frac{1}{48}t^4 - \frac{1}{16}t^2.$$

For the exponential part, try

$$Y_2 = De^t,$$

since $r = 1$ is not a root of the characteristic equation. Then

$$Y_2^{(4)} - 4Y_2'' = (1 - 4)De^t = -3De^t.$$

Matching with e^t gives

$$D = -\frac{1}{3} \Longrightarrow Y_2 = -\frac{1}{3}e^t \Longrightarrow Y = -\frac{1}{48}t^4 - \frac{1}{16}t^2 - \frac{1}{3}e^t.$$

Hence the general solution is

$$\boxed{y(t) = c_1 + c_2 t + c_3 e^{2t} + c_4 e^{-2t} - \frac{1}{48}t^4 - \frac{1}{16}t^2 - \frac{1}{3}e^t}$$

(ii) $y^{(4)} + y''' = \sin 2t$

First solve the homogeneous equation

$$y^{(4)} + y''' = 0.$$

Its characteristic equation is

$$r^4 + r^3 = 0 \quad \Longrightarrow \quad r^3(r + 1) = 0.$$

So the roots are

$$r = 0, 0, 0, -1.$$

Thus $r = 0$ has multiplicity 3, and the homogeneous solution is

$$y_c(t) = c_1 + c_2 t + c_3 t^2 + c_4 e^{-t}.$$

For

$$y^{(4)} + y''' = \sin 2t,$$

Use the standard trial form

$$Y = A\sin 2t + B\cos 2t,$$

Compute derivatives:

$$Y' = 2A\cos 2t - 2B\sin 2t,$$
$$Y'' = -4A\sin 2t - 4B\cos 2t,$$
$$Y''' = -8A\cos 2t + 8B\sin 2t,$$
$$Y^{(4)} = 16A\sin 2t + 16B\cos 2t.$$

Therefore,

$$Y^{(4)} + Y''' = (16A + 8B)\sin 2t + (16B - 8A)\cos 2t.$$

Match coefficients with $\sin 2t$:

$$16A + 8B = 1, \qquad 16B - 8A = 0.$$

From the second equation,

$$B = \frac{A}{2}.$$

Substitute into the first:

$$16A + 8\left(\frac{A}{2}\right) = 16A + 4A = 20A = 1,$$

so

$$A = \frac{1}{20}, \qquad B = \frac{1}{40}.$$

Hence

$$Y = \frac{1}{20}\sin 2t + \frac{1}{40}\cos 2t.$$

Therefore the general solution is

$$\boxed{y(t) = c_1 + c_2 t + c_3 t^2 + c_4 e^{-t} + \frac{1}{20}\sin 2t + \frac{1}{40}\cos 2t}$$

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(i) $y''' - y'' + y' - y = \sec t, \quad -\pi/2 < t < \pi/2$

1. Complementary Solution (y_c):

The characteristic equation is $r^3 - r^2 + r - 1 = 0$, which factors as $(r^2 + 1)(r - 1) = 0$. The roots are $r_1 = 1, r_2 = i, r_3 = -i$, giving the fundamental set of solutions:

- $y_1(t) = e^t$
- $y_2(t) = \cos t$
- $y_3(t) = \sin t$ Thus, $y_c(t) = c_1 e^t + c_2 \cos t + c_3 \sin t$.

2. Particular Solution (Y) via Variation of Parameters:

Let $Y(t) = u_1(t)y_1 + u_2(t)y_2 + u_3(t)y_3$. The Wronskian of the fundamental set is:

$$W(e^t, \cos t, \sin t) = \det \begin{pmatrix} e^t & \cos t & \sin t \\ e^t & -\sin t & \cos t \\ e^t & -\cos t & -\sin t \end{pmatrix} = 2e^t$$

Using Cramer's rule, find the derivatives of the parameters:

- $u_1'(t) = \frac{\det M_1 \cdot \sec t}{W} = \frac{1 \cdot \sec t}{2e^t} = \frac{1}{2}e^{-t}\sec t$
- $u_2'(t) = \frac{\det M_2 \cdot \sec t}{W} = \frac{e^t(\sin t - \cos t)\sec t}{2e^t} = \frac{1}{2}(\tan t - 1)$
- $u_3'(t) = \frac{\det M_3 \cdot \sec t}{W} = \frac{-e^t(\sin t + \cos t)\sec t}{2e^t} = -\frac{1}{2}(\tan t + 1)$

General Solution:

$$y(t) = c_1 e^t + c_2 \cos t + c_3 \sin t + \frac{e^t}{2} \int e^{-t} \sec t \, dt + \frac{\cos t}{2} \int (\tan t - 1) \, dt - \frac{\sin t}{2} \int (\tan t + 1) \, dt$$

(ii) $y''' - y' = \csc t, \quad 0 < t < \pi$

1. Complementary Solution (y_c):

The characteristic equation is $r^3 - r = 0$, which factors as $r(r^2 - 1) = 0$. The roots are $r_1 = 0, r_2 = 1, r_3 = -1$. The fundamental set of solutions is:

- $y_1(t) = 1$
- $y_2(t) = e^t$
- $y_3(t) = e^{-t}$ Thus, $y_c(t) = c_1 + c_2 e^t + c_3 e^{-t}$.

2. Particular Solution (Y) via Variation of Parameters:

The Wronskian of the fundamental set is:

$$W(1, e^t, e^{-t}) = \det \begin{pmatrix} 1 & e^t & e^{-t} \\ 0 & e^t & -e^{-t} \\ 0 & e^t & e^{-t} \end{pmatrix} = 2$$

Using Cramer's rule, the derivatives of the parameters are:

- $u_1'(t) = \frac{\det M_1 \cdot \csc t}{W} = \frac{-2 \csc t}{2} = -\csc t$
- $u_2'(t) = \frac{\det M_2 \cdot \csc t}{W} = \frac{e^{-t} \csc t}{2}$
- $u_3'(t) = \frac{\det M_3 \cdot \csc t}{W} = \frac{e^t \csc t}{2}$

General Solution:

$$y(t) = c_1 + c_2 e^t + c_3 e^{-t} - \int \csc t \, dt + \frac{e^t}{2} \int e^{-t} \csc t \, dt + \frac{e^{-t}}{2} \int e^t \csc t \, dt$$

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Since the forcing term is $2x^4$, and x^4 is not among the homogeneous solutions x, x^2, x^{-1}, it is natural to try

$$y_p = Ax^4.$$

Now compute:

$$y_p' = 4Ax^3, \qquad y_p'' = 12Ax^2, \qquad y_p''' = 24Ax.$$

Substitute into

$$x^3 y''' + x^2 y'' - 2xy' + 2y = 2x^4 :$$

$$x^3(24Ax) + x^2(12Ax^2) - 2x(4Ax^3) + 2(Ax^4) = 2x^4.$$

So

$$(24A + 12A - 8A + 2A)x^4 = 2x^4,$$

hence

$$30A = 2 \quad \Rightarrow \quad A = \frac{1}{15}.$$

Therefore, a particular solution is

$$\boxed{y_p(x) = \frac{1}{15} x^4.}$$

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Let

$$x_1(t) = u(t), \qquad x_2(t) = u'(t).$$

Then

$$x_1'(t) = x_2(t).$$

From

$$u'' + p(t)u' + q(t)u = g(t),$$

we get

$$u'' = g(t) - p(t)u' - q(t)u,$$

so

$$x_2'(t) = g(t) - p(t)x_2(t) - q(t)x_1(t).$$

Hence the given second-order initial value problem is transformed into the first-order system (inn matrix form)

$$\boxed{\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -q(t) & -p(t) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ g(t) \end{pmatrix}, \qquad \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} u_0 \\ u_0' \end{pmatrix}.}$$

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(i) $\boldsymbol{x}' = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \boldsymbol{x}$

1. Find the Eigenvalues:

Solve the characteristic equation $\det(\mathbf{A} - r\mathbf{I}) = 0$:

$$\begin{vmatrix} 1-r & 2 \\ 4 & 3-r \end{vmatrix} = (1-r)(3-r) - 8 = r^2 - 4r + 3 - 8 = r^2 - 4r - 5 = 0$$

$$(r-5)(r+1) = 0 \implies r_1 = 5, \quad r_2 = -1$$

The eigenvalues are real and distinct.

2. Find the Eigenvectors:

- **For $r_1 = 5$:**

$$(\mathbf{A} - 5\mathbf{I})\boldsymbol{\xi} = \mathbf{0} \implies \begin{bmatrix} -4 & 2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies -4\xi_1 + 2\xi_2 = 0 \implies \xi_2 = 2\xi_1$$

  Choose $\boldsymbol{\xi}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

- **For $r_2 = -1$:**

$$(\mathbf{A} + \mathbf{I})\boldsymbol{\xi} = \mathbf{0} \implies \begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies 2\xi_1 + 2\xi_2 = 0 \implies \xi_1 = -\xi_2$$

  Choose $\boldsymbol{\xi}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

3. General Solution:

$$\boxed{\boldsymbol{x}(t) = c_1 e^{5t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}}$$

(ii) $\boldsymbol{x}' = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix} \boldsymbol{x}$

1. Find the Eigenvalues:

$$\begin{vmatrix} 2-r & -1 \\ 1 & 4-r \end{vmatrix} = (2-r)(4-r) + 1 = r^2 - 6r + 8 + 1 = r^2 - 6r + 9 = 0$$

$$(r-3)^2 = 0 \implies r = 3 \quad \text{(Algebraic multiplicity 2)}$$

2. Find the Eigenvector and Generalized Eigenvector:

- **Eigenvector $\boldsymbol{\xi}$:**

$$(\mathbf{A} - 3\mathbf{I})\boldsymbol{\xi} = \mathbf{0} \implies \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \xi_1 + \xi_2 = 0$$

  Choose $\boldsymbol{\xi} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

- **Generalized Eigenvector $\boldsymbol{\eta}$:** Solve $(\mathbf{A} - 3\mathbf{I})\boldsymbol{\eta} = \boldsymbol{\xi}$:

$$\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies -\eta_1 - \eta_2 = 1 \implies \eta_1 + \eta_2 = -1$$

Choose $\eta_2 = 0$, then $\eta_1 = -1$. So, $\boldsymbol{\eta} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$.

3. General Solution:

$$\boxed{\boldsymbol{x}(t) = c_1 e^{3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + c_2 e^{3t} \left(t \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \end{bmatrix} \right)}$$

(iii) $\boldsymbol{x}' = \begin{bmatrix} 1 & -5 \\ 1 & -3 \end{bmatrix} \boldsymbol{x}, \quad \boldsymbol{x}(0) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

1. Find the Eigenvalues:

$$\begin{vmatrix} 1 - r & -5 \\ 1 & -3 - r \end{vmatrix} = (1 - r)(-3 - r) + 5 = r^2 + 2r + 2 = 0$$

$$r = \frac{-2 \pm \sqrt{4 - 8}}{2} = -1 \pm i$$

2. Find the Eigenvector for $r = -1 + i$:

$$(\mathbf{A} - (-1 + i)\mathbf{I})\boldsymbol{\xi} = \mathbf{0} \implies \begin{bmatrix} 2 - i & -5 \\ 1 & -2 - i \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the second row: $\xi_1 = (2 + i)\xi_2$. Choose $\xi_2 = 1 \implies \xi_1 = 2 + i$.

$$\boldsymbol{\xi} = \begin{bmatrix} 2 + i \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + i \begin{bmatrix} 1 \\ 0 \end{bmatrix} \implies \mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

3. Construct the Real-Valued General Solution:

$$\mathbf{y}_1(t) = e^{-t}(\mathbf{u}\cos t - \mathbf{v}\sin t) = e^{-t} \begin{bmatrix} 2\cos t - \sin t \\ \cos t \end{bmatrix}$$

$$\mathbf{y}_2(t) = e^{-t}(\mathbf{u}\sin t + \mathbf{v}\cos t) = e^{-t} \begin{bmatrix} 2\sin t + \cos t \\ \sin t \end{bmatrix}$$

$\boldsymbol{x}(t) = c_1\mathbf{y}_1(t) + c_2\mathbf{y}_2(t)$.

4. Apply Initial Condition $\boldsymbol{x}(0) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$:

$$\boldsymbol{x}(0) = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \implies c_1 = 1, \quad c_2 = 0$$

Particular Solution:

$$\boxed{\boldsymbol{x}(t) = e^{-t} \begin{bmatrix} 2\cos t - \sin t \\ \cos t \end{bmatrix}}$$